

● EASY

● ADVANCED MATH

● ANSWER KEY

---

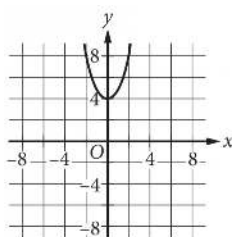
# Nonlinear functions

# 06

10 answers · Rationale-rich · Teach from doc

**Q1****3**

The correct answer is 3. The value of  $f(0)$  is the value of  $y$  on the graph of  $y = f(x)$  in the  $xy$ -plane that corresponds with  $x = 0$ . It's given that the value of  $f(0)$  is an integer. For the graph of  $y = f(x)$  shown, when  $x = 0$ , the corresponding integer value of  $y$  is 3. Therefore, the value of  $f(0)$  is 3.

**Q2****D****D.**

Choice D is correct. For the quadratic function  $f(x) = x^2 + 4$ , the vertex of the graph is  $(0, 4)$ , and because the  $x^2$  term is positive, the vertex is the minimum of the function. Choice D is the only option that meets these conditions.

Choices A, B, and C are incorrect. The vertex of each of these graphs doesn't correspond to the minimum of the given function.

**Q3****A**

- A. 5 years after the company started selling jewelry online, its predicted annual revenue is approximately 648,650 dollars.
- 

Choice A is correct. It's given that the function  $f$  gives a company's predicted annual revenue, in dollars,  $x$  years after the company started selling jewelry online. Since the value of  $f(5)$  is the value of  $f(x)$  when  $x = 5$ , it follows that " $f(5)$  is approximately equal to 648,650" means that  $f(x)$  is approximately equal to 648,650 when  $x = 5$ . Therefore, the best interpretation of the given statement is that 5 years after the company started selling jewelry online, its predicted annual revenue is approximately 648,650 dollars.

Choice B is incorrect. The function  $f$  gives the predicted annual revenue, not the predicted increase in annual revenue.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. In the given function,  $x$  represents the number of years after the company started selling jewelry online, not the percent increase in revenue from the previous year.

---

**Q4****D**

- D. 320
- 

Choice D is correct. It's given that the function  $f$  is defined by  $fx = 5x^2$ . Substituting 8 for  $x$  in  $fx = 5x^2$  yields  $f8 = 58^2$ , which is equivalent to  $f8 = 564$ , or  $f8 = 320$ . Therefore, the value of  $f8$  is 320.

Choice A is incorrect. This is the value of  $f8$  if  $fx = 5x$ .

Choice B is incorrect. This is the value of  $f8$  if  $fx = 5x + 2$ .

Choice C is incorrect. This is the value of  $f8$  if  $fx = 5x2$ .

---

**Q5**

**D**

D. 4, 0

---

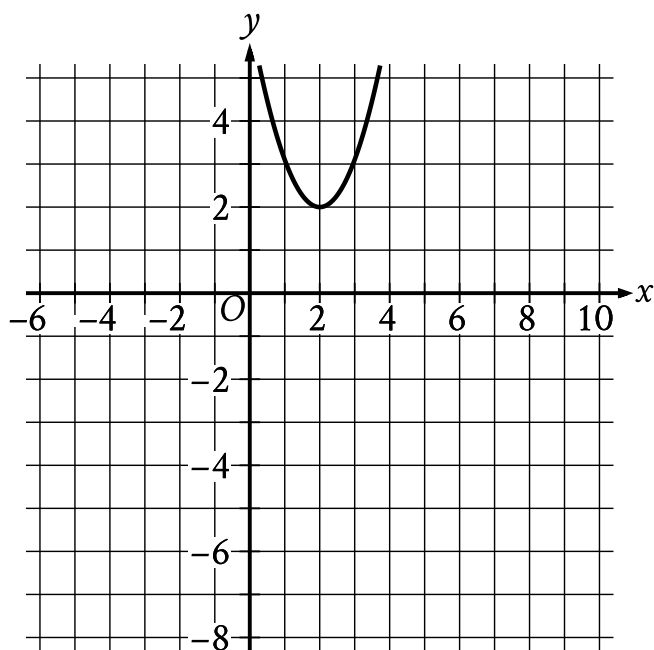
Choice D is correct. The  $x$ -intercept of the graph shown is the point  $x, y$  on the graph where  $y = 0$ . At  $y = 0$ , the corresponding value of  $x$  is 4. Therefore, the  $x$ -intercept of the graph shown is 4, 0.

Choice A is incorrect. This is the  $x$ -intercept of a graph in the  $xy$ -plane that intersects the  $x$ -axis at  $x = -5$ , not  $x = 4$ .

Choice B is incorrect. This is the  $x$ -intercept of a graph in the  $xy$ -plane that intersects the  $x$ -axis at  $x = 5$ , not  $x = 4$ .

Choice C is incorrect. This is the  $x$ -intercept of a graph in the  $xy$ -plane that intersects the  $x$ -axis at  $x = -4$ , not  $x = 4$ .

---

**Q6****A****A.**

- The parabola opens upward.
- The vertex is at the point (2, 2).
- The parabola passes through the following points:
  - (1, 3)
  - (2, 2)
  - (3, 3)

Choice A is correct. When a graph is translated up 4 units, each point on the resulting graph is 4 units above the point on the original graph. In other words, the  $y$ -value of each point on the graph increases by 4. The graph shown passes through the points (1, -1), (2, -2), and (3, -1). It follows that when the graph shown is translated up 4 units, the resulting graph will pass through the points (1, -1 + 4), (2, -2 + 4), and (3, -1 + 4). These points are (1, 3), (2, 2), and (3, 3), respectively. Of the given choices, only the graph in choice A passes through the points (1, 3), (2, 2), and (3, 3).

Choice B is incorrect. This is the result of translating the graph down, rather than up, 4 units.

Choice C is incorrect. This is the result of translating the graph left, rather than up, 4 units.

Choice D is incorrect. This is the result of translating the graph right, rather than up, 4 units.

---

**Q7**

**C**

**C.** 1,250

---

Choice C is correct. Each point  $(x, y)$  on the graph represents the elevation  $x$ , in meters, and the corresponding measure of plant diversity  $y$  in a region of Switzerland. Therefore, the point on the graph with the greatest  $y$ -coordinate represents the location that has the greatest measure of plant diversity in the region. The greatest  $y$ -coordinate of any point on the graph is approximately 13,500. The  $x$ -coordinate of that point is approximately 1,250. Therefore, the closest elevation at which the plant diversity is the greatest is 1,250 meters.

Choice A is incorrect. This value is closest to the greatest  $y$ -coordinate of any point on the graph and therefore represents the greatest measure of plant diversity, not the elevation where the greatest measure of plant diversity occurs. Choice B is incorrect. At an elevation of 3,000 meters the measure of plant diversity is approximately 4,000. Because there are points on the graph with greater  $y$ -coordinates, 4,000 can't be the greatest measure of plant diversity, and 3,000 meters isn't the elevation at which the greatest measure of plant diversity occurs. Choice D is incorrect. At an elevation of 250 meters, the measure of plant diversity is approximately 11,000. Because there are points on the graph with greater  $y$ -coordinates, 11,000 can't be the greatest measure of plant diversity and 250 meters isn't the elevation at which the greatest measure of plant diversity occurs.

---

**Q8****C****C.** 750

Choice C is correct. Adding the like terms  $x$  and  $0.25x$  yields the equation  $f(x) = (1.25x)(50 - x)$ . Substituting 20 for  $x$  yields  $f(20) = (1.25(20))(50 - 20)$ . The product  $1.25(20)$  is equal to 25, and the difference  $50 - 20$  is equal to 30. Substituting these values in the given equation gives  $f(20) = (25)(30)$ , and multiplying 25 by 30 results in  $f(20) = 750$ .

Choices A, B, and D are incorrect and may result from conceptual or computational errors when finding the value of  $f(20)$ .

**Q9****D****D.** 0, 8

Choice D is correct. The  $y$ -intercept of a graph in the  $xy$ -plane is the point at which the graph crosses the  $y$ -axis. The graph shown crosses the  $y$ -axis at the point 0, 8. Therefore, the  $y$ -intercept of the graph shown is 0, 8.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

## Q10

C

C.  $p = 240\left(\frac{1}{2}\right)^r$

Choice C is correct. It's given that one of the two players in a match is eliminated during each round. Therefore, the number of players decreases by half at the end of each round, and so the number of players eliminated at the end of a round also decreases by half. A decreasing exponential equation can be written in the form  $p = a(1 - b)^r$ , where  $b$  is the value of the decrease each round,  $r$ , and  $a$  is the value of  $p$ , the number of players, when  $r = 0$ . It's given that there are 240 players in the tennis competition. Therefore,  $a = 240$ . Since the number of players eliminated at the end of a round decreases by half, it follows that  $b = \frac{1}{2}$ . Substituting 240 for  $a$  and  $\frac{1}{2}$  for  $b$  in the equation  $p = a(1 - b)^r$  yields  $p = 240\left(1 - \frac{1}{2}\right)^r$ , or  $p = 240\left(\frac{1}{2}\right)^r$ .

Choice A is incorrect. This equation gives the number of players eliminated at the end of round  $r$  for a competition in which there are 15, not 240, players.

Choice B is incorrect. This equation gives the number of players eliminated at the end of round  $r$  for a competition in which there are 15, not 240, players, and for which the number of players doubles, rather than decreases by half, after each round.

Choice D is incorrect. This equation gives the number of players eliminated at the end of round  $r$  for a competition in which the number of players doubles, rather than decreases by half, after each round.

### Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank